

# Variance Reduction in Monte Carlo Capacitance Extraction

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**Abstract**—In this article we address efficiency issues in implementation of Monte Carlo algorithm for 3D capacitance extraction. Error bounds in statistical capacitance estimation are discussed. Methods to tighten them through variance reduction techniques are detailed. Sample values in implementation of Monte Carlo algorithm is completely determined by the first hop in random walk. This in turn facilitates application of variance reduction techniques like *importance sampling* and *stratified sampling* to be used effectively. Experimental results indicate average speedup of 16X in simple uniform dielectric technologies, 7.3X in technologies with layers of dielectrics and 4.6X in technologies having conformal dielectrics.

## I. INTRODUCTION

Interconnect capacitances, both self and coupling, are of immense importance in sub-quarter micron layouts. Their accurate computation is key to the success of any subsequent timing or signal integrity analyses. There are various numerical techniques like Boundary Element Method (BEM), Finite Element Method (FEM), Monte Carlo Integration, etc. which can compute these capacitances at different levels of accuracy. BEM and FEM discretise boundary and volume into elements, and write electrostatic equations at each element. The resulting linear equations are solved by efficient techniques. Monte Carlo methods compute capacitance by evaluating the charge integral by statistical sampling techniques. The statistical sampling is implemented through random walk process.

Efficient implementation of BEM using method of moments and sparsification of dense matrices is given in [1] and [2]. The basic algorithm is extended to handle multiple dielectrics in [3]. Further extensions are given to facilitate efficient handling of conformal dielectrics in [4] and [5]. In [6] the method is extended to handle high permittivity ratios of dielectric layers. These are fast multipole methods. Statistical methods use Monte Carlo techniques to evaluate the charge integral. Instead of discretising volume or boundaries they evaluate charge integral by efficiently sampling of the integrand in the integral. A good survey of capacitance extraction methods is given in [12].

Among all these methods Monte Carlo methods have emerged as most successful in computing self and coupling capacitances for interconnects with sign-off accuracy. One of the main advantage being their scalability to large designs and higher accuracy due to statistical error cancellation. Furthermore, these methods are best suited for parallel implementations. Foundations of these methods stem from the work of Muller [7] which describes Monte Carlo methods to solve Dirichlet problems. The first algorithmic description of the Monte Carlo methods for capacitance extraction problems is given in the pioneering work of Le Coz and Iverson [8]. Statistical samples for the integrand in the charge integral are evaluated through floating random walks in the space around the conductors. Efficient extensions of this algorithm to handle multiple dielectrics are given by Schlott [9] and Le Coz and Iverson [10], [11]. The capacitance numbers computed by these methods have an associated statistical uncertainty which is a good indicator of the error in capacitance values. There is a usual tradeoff between accuracy and performance.

Unlike BEM based methods, not much has been published towards efficient implementation of random walk methods. Though a parallel implementation of the random walks is detailed in [13], not much is published about possible improvements for Monte Carlo integration of charge integral. It is well known that Monte Carlo implementations can be made efficient by incorporating various *variance reduction* methods

(see [14]). In this article, we closely study the random walk process and suggest a couple of variance reduction methods which improve the capacitance extraction performance by up-to 16X. In Section II-A we briefly rewrite the fundamental electrostatic equations used in capacitance extraction. In Section II-B we describe basic Monte Carlo process to solve integral equations statistically. Later, in Section II-C we apply the Monte Carlo integration technique to solve electrostatic equations through random walks. In Section III we study a few of the standard variance reduction techniques whose application is beneficial in our case. In Section IV we present a modified random walk algorithm which uses these variance reduction techniques. In Section V we present a few of the experimental results to quantify the benefits of using variance reduction methods. Finally in Section VI, we list our concluding remarks.

## II. MONTE CARLO CAPACITANCE EXTRACTION

### A. Fundamental Electrostatic Equations

In VLSI design, capacitance extraction problem is described as follows. Given geometric description of nets in the design and technology details, like relative permittivity and thickness of various inter layer dielectric and conformal dielectric layers, it is desired to compute self and coupling capacitances associated with user specified nets. The formulation of this problem results into an electrical system with  $n$  nets or conductors. In an electrical system with  $n$  conductors the total charge  $Q_i$  on conductor  $i$  can be expressed in terms of the mutual capacitance  $C_{ij}$  between conductor  $i$  and  $j$  and the potential vector  $(v_1, \dots, v_n)$ .

$$Q_i = \sum_{j=1, j \neq i}^n C_{ij}(v_i - v_j). \quad (1)$$

If we keep conductor  $i$  at 0 volt and all the other conductors at 1 volt, we can compute the total capacitance (also called *self capacitance*) of conductor  $i$  and the mutual capacitances (also called *coupling capacitances*) between conductor  $i$  and the other conductors in the system. The self capacitance is equal to the total charge on  $i$  and the coupling capacitances are computed by linear separation of  $Q_i$  in Equation 1.

Basic electrostatic equations used for evaluation of charge  $Q$  are given below.<sup>1</sup>

$$Q = - \int_{\Gamma} dp \epsilon_p \hat{\mathbf{E}}(p) \cdot \hat{\boldsymbol{\eta}}(p) \quad (2)$$

$$\hat{\mathbf{E}}(p) = \int_{C_p} dp_1 \hat{\mathbf{G}}_p(p_1) \phi(p_1) \quad (3)$$

$$\phi(p_1) = \int_{C_{p_1}} dp_2 G_{p_1}(p_2) \phi(p_2) \quad (4)$$

According to Gauss's law, total charge  $Q$  stored on a conductor in a multi-conductor system is evaluated by integrating dot product of electric field  $\hat{\mathbf{E}}(\cdot)$  and unit normal vector  $\hat{\boldsymbol{\eta}}(\cdot)$  on a surface  $\Gamma$  around the conductor. This surface is often called *Gaussian surface*. By definition, Gaussian surface *should* completely enclose the conductor on

<sup>1</sup> For notational simplicity we represent surface integral by simple integral sign  $\int$ .

which total charge is being computed and *should not* contain any other conductor in the system. Given geometric description of the conductor system, a Gaussian surface can be constructed, and unit normal vectors can be computed at every point on the surface. However, the electric field  $\hat{\mathbf{E}}(p)$  at points  $p$  on Gaussian surface need to be evaluated.  $\hat{\mathbf{E}}(p)$  can be computed in terms of electric potentials around a conformal boundary around the point  $p$ . For algorithmic efficiency this boundary is taken as a cube  $C_p$  centered around point  $p$ . The field is related to boundary potentials through a vector Green's function  $\hat{\mathbf{G}}_p(\cdot)$  as shown in Equation 3. Potential at points  $p_1$  on surface of this cube can again be expressed in terms of boundary potentials around another cube  $C_{p_1}$  centered around point  $p_1$ . The relationship involves a scalar Green's function  $G_{p_1}(\cdot)$  as shown in Equation 4. These vector and scalar Green's functions can be easily computed once the internal dielectric composition of cube  $C_p$  or  $C_{p_1}$  is specified.

Let us now look at a computation scheme for evaluating  $Q$ . We use the algorithm identical to the one given in [8]. The dot product  $\hat{\mathbf{E}}(p) \cdot \hat{\eta}(p)$  can be represented as,  $E_x(p)\eta_x(p) + E_y(p)\eta_y(p) + E_z(p)\eta_z(p)$ . Also, since  $\hat{\mathbf{E}}(\cdot)$  is a three dimensional vector, Green's function  $\hat{\mathbf{G}}_p(\cdot)$  also has three components. The dot product,  $\hat{\mathbf{G}}_p(p_1) \cdot \hat{\eta}(p)$  can also be represented as  $G_p^x(p_1)\eta_x(p) + G_p^y(p_1)\eta_y(p) + G_p^z(p_1)\eta_z(p)$ , where  $G_p^x(\cdot)$ ,  $G_p^y(\cdot)$  and  $G_p^z(\cdot)$  can be thought of as scalar Green's functions or component functions for vector function  $\hat{\mathbf{G}}_p(\cdot)$ . Let us define a weight function  $w_p(\cdot)$  at point  $p$  on Gaussian surface  $\Gamma$ .

$$w_p(p_1) = -\epsilon_p \{ G_p^x(p_1)\eta_x(p) + G_p^y(p_1)\eta_y(p) + G_p^z(p_1)\eta_z(p) \}. \quad (5)$$

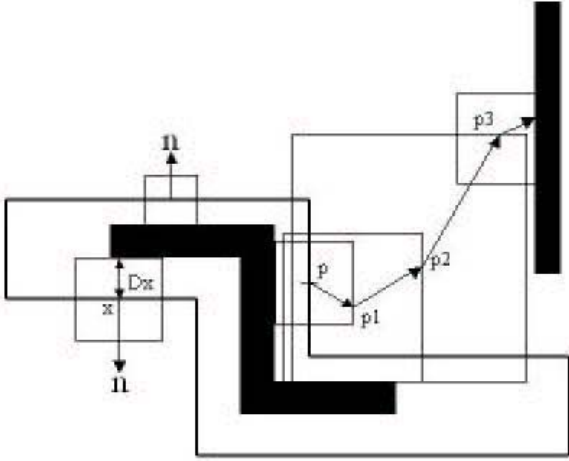


Fig. 1. Illustration of random walk process.

Rewriting Equation 2 into its component form, we get,

$$Q = -\epsilon_p \int_{\Gamma} dp \{ E_x(p) \cdot \eta_x(p) + E_y(p) \cdot \eta_y(p) + E_z(p) \cdot \eta_z(p) \} \quad (6)$$

Using the weight function  $w_p(\cdot)$ ,

$$Q = \int_{\Gamma} dp \left\{ \int_{C_p} dp_1 w_p(p_1) \phi(p_1) \right\}. \quad (7)$$

The domain for inner surface integral is a maximal conductor free cube  $C_p$  centered at point  $p$ . It can be decomposed into two parts as

$$C_p = C_p^s \cup C_p^c, \quad (8)$$

where  $C_p^c$  is part of cube's surface which is in contact with conductors and  $C_p^s$  is remaining part of cube's surface which is in dielectric space.

Since all other conductors are kept at unit potential, we have potential at any point on  $C_p^c = 1$  volt. Using this cube decomposition and Equation 4 in Equation 7 we get,

$$\begin{aligned} Q &= \int_{\Gamma} dp \left\{ \int_{C_p^c} dp_1 w_p(p_1) \phi(p_1) + \int_{C_p^s} dp_1 w_p(p_1) \phi(p_1) \right\} \quad (9) \\ &= \int_{\Gamma} dp \left\{ \int_{C_p^c} dp_1 w_p(p_1) + \int_{C_p^s} dp_1 w_p(p_1) \right. \\ &\quad \left. \left\{ \int_{C_{p_1}^c} dp_2 G_{p_1}(p_2) \phi(p_2) + \int_{C_{p_1}^s} dp_2 G_{p_1}(p_2) \phi(p_2) \right\} \right\} \quad (10) \\ &= \int_{\Gamma} dp \left\{ \int_{C_p^c} dp_1 w_p(p_1) + \int_{C_p^s} dp_1 w_p(p_1) \right. \\ &\quad \left\{ \int_{C_{p_1}^c} dp_2 G_{p_1}(p_2) + \int_{C_{p_1}^s} dp_2 G_{p_1}(p_2) \right. \\ &\quad \left. \left\{ \int_{C_{p_2}^c} dp_3 G_{p_2}(p_3) \phi(p_3) + \int_{C_{p_2}^s} dp_3 G_{p_2}(p_3) \dots \right\} \right\} \quad (11) \end{aligned}$$

### B. Monte Carlo Integration

We now need to evaluate the charge integral of Equation 11. This can be done by Monte Carlo integration [14]. Before we go into specifics of charge integral let us consider Monte Carlo evaluation in a general setting. Let us integrate a multi-dimensional function  $\mathbf{f}(\cdot)$  over a domain  $\Omega$ .

$$\theta = \int_{\Omega} \mathbf{f}(x) dx \quad (12)$$

If  $\int_{\Omega} dx = k$  then we pick  $N$  random samples  $x_i$  uniformly from  $\Omega$  and evaluate an estimate  $\theta_N$  for  $\theta$  as follows,

$$\theta_N = k \frac{\sum_{i=1}^N \mathbf{f}(x_i)}{N}. \quad (13)$$

The variance of the sample space is

$$\sigma_N^2 = \frac{k^2 \sum_{i=1}^N (\mathbf{f}(x_i))^2 - N \theta_N^2}{N-1} \quad (14)$$

The standard error  $e_N$  associated with the estimate  $\theta_N$  is

$$e_N = \frac{\sigma_N}{\sqrt{N}} \quad (15)$$

It is evident that the standard error drops as  $1/\sqrt{N}$ , i.e., to say that in order to reduce the error by a factor of  $2X$  we need to consider a sample space of size  $4X$ .

We use a slightly modified version of Monte Carlo integration as well. Let us consider integrating the following function,

$$\theta = \int_{\Omega} \mathbf{S}(x) \mathbf{f}(x) dx, \quad (16)$$

where  $\int_{\Omega} \mathbf{S}(x) dx = 1$ . If we pick  $x_i$  uniformly from  $\Omega$  then  $\theta_N = k \sum_{i=1}^N \mathbf{S}(x_i) \mathbf{f}(x_i) / N$ . However, if we pick  $x_i$  from  $\Omega$  with probability  $\mathbf{S}(x_i)$ , then  $\theta_N = \sum_{i=1}^N \mathbf{f}(x_i) / N$ . Both result in the same estimate  $\theta_N$ .

### C. Random Walk Process

Let us now apply Monte Carlo integration procedure outlined above to evaluate the charge integral given in Equation 11. If  $\int_{\Gamma} dp = A_{\Gamma}$ , then we compute an estimate  $Q_N$  of  $Q$  as,

$$Q_N = A_{\Gamma} \frac{\sum_{i=1}^N q_i}{N}, \quad (17)$$

where  $q_i$  is a sample of integrand in Equation 11 (equivalently Equation 2). In other words,  $q_i$  is the value of dot product of  $\hat{\mathbf{E}}$  and  $\hat{\boldsymbol{\eta}}$  vector at a point  $p_i$  on surface  $\Gamma$ . Given a point  $p_i$  and the description of surface  $\Gamma$ , the normal vector  $\hat{\boldsymbol{\eta}}$  is a known entity. However, field  $\hat{\mathbf{E}}$  at  $p_i$  needs to be estimated. We estimate it by sampling a point on a conductor free cube  $C_{p_i}$  centered at  $p_i$ . This is done by doing Monte Carlo integration of Equation 3. We pick a point  $p_j$  uniformly from the surface of  $C_{p_i}$ . If surface area of cube  $C_{p_i} = A_{C_i}$ , then  $q_i$  is obtained as,

$$q_i = -A_{C_i} \cdot \epsilon_{p_i} \hat{\mathbf{G}}_{p_i}(p_j) \cdot \hat{\boldsymbol{\eta}}_{p_i} \phi(p_j) \quad (18)$$

Once again, to evaluate the above quantity we need to evaluate  $\phi(p_j)$ . If point  $p_j$  is on the surface of any conductor, then we know that  $\phi(p_j) = 1$  volt, otherwise we need to estimate it. A random walk starting from point  $p$  on a Gaussian surface and then going through points  $p_1, p_2$  and  $p_3$  before terminating on another conductor is shown in Figure 1.

We do a slightly different Monte Carlo integration of Equation 4 towards this. Let  $G_{p_j}(p_k)$  be the Potential Green's function of the dielectric cube centered at  $p_j$ . We know that  $\int_{C_{p_j}} G_{p_j}(p) dp = 1$ . In other words, if  $\wp$  is the set of points on the surface of  $C_{p_j}$ , then we know that,  $\sum_{p_k \in \wp} G_{p_j}(p_k) = 1$ . Thus  $G_{p_j}$  can be used as a probability density function (pdf). We pick a point  $p_k$  on the surface of  $C_{p_j}$  with probability  $^2 G_{p_j}(p_k)$ . Thus the sample of integrand in Equation 4 is always of value  $\phi(p_k)$ . If  $p_k$  is on surface of a conductor then  $\phi(p_k) = 1$ , otherwise we once again use Monte Carlo integration of Equation 4 to evaluate  $\phi(p_k)$ .

With arguments in above paragraphs, each random walk results into a sample of value  $q_i$  which is completely determined by the dot-product evaluated in first step. In particular,

$$q_i = A_{C_i} \cdot \epsilon_{p_i} \hat{\mathbf{G}}_{p_i}(p_j) \cdot \hat{\boldsymbol{\eta}}_{p_i} \quad (19)$$

### III. OVERVIEW OF VARIANCE REDUCTION TECHNIQUES

The Monte Carlo procedure sketched in Section II-B is also called 'crude Monte Carlo'[14]. It does not exploit structure and physical properties of the integral being evaluated. There are various sampling schemes using which the variance of the estimates can be reduced. A detailed survey of them can be obtained from standard texts on Monte Carlo integration [14], [15], [16], [17]. These variance reduction techniques, though easily stated mathematically, are difficult to apply to particular instances of physical problems. One needs to understand properties and structure of the integrand function and use them effectively to identify and use suitable variance reduction techniques. In this section we look at a few of the standard techniques whose application to integration of Equation 2 results into good variance reduction.

#### A. Domain Selection

The charge integral of Equation 2 can be evaluated by choosing any integration boundary which satisfies the requirements of a Gaussian surface. All such boundaries result into same total charge. However, the convergence behavior of the Monte Carlo scheme is different for different boundaries. In Figure 2 we plot convergence behavior for self capacitance of a net in standard cell design (total 75 rectilinear nets, 1667 3-D boxes in geometric description of design, 4 metal layer technology). We choose an integration boundary very close to the net's surface. Since the electric field is very ill-behaved near the conductor surface we get large standard error in our estimation. In Figure 2 we plot the capacitance value with associated errorbars, as computed by standard error. We know that electric field is very well behaved at points equi-distant from conductors. We chose an integration boundary around the net being extracted in such a way that points on it are equi-distant from the underlying net and from other adjacent nets. The

convergence behavior is shown in Figure 3. We keep the number of samples same as that in Figure 2 for easy comparison. However, note that the Y-axis is different in these figures. The error bars are much smaller in Figure 3 indicating that the convergence behavior is much better with equi-distant Gaussian surface.<sup>3</sup> In our implementation, and for all the results reported in this paper, we always use equi-distant Gaussian surface.

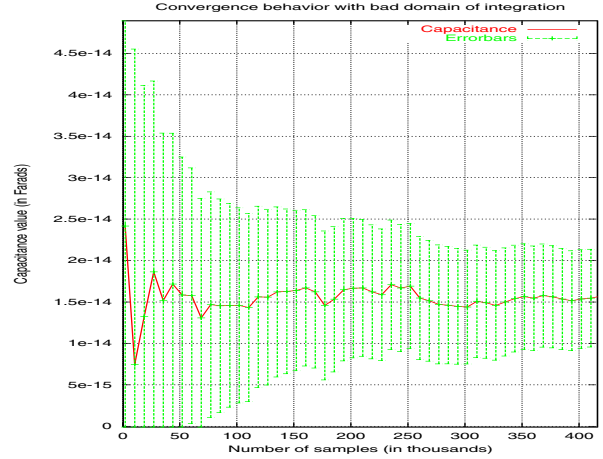


Fig. 2. Convergence behavior of a self cap with bad domain of integration.

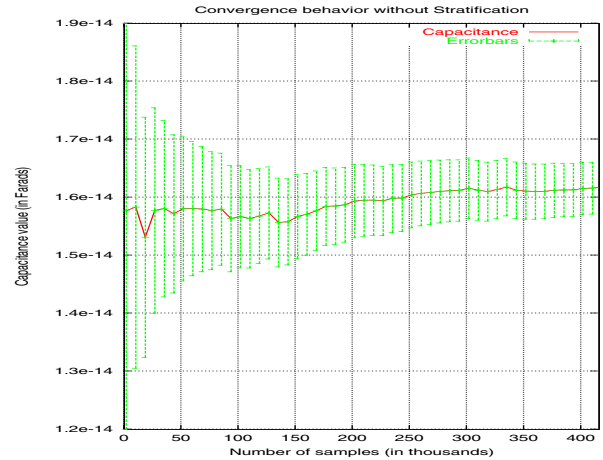


Fig. 3. Convergence behavior of a self cap without any stratification.

#### B. Importance Sampling

If we can compute a positive valued function  $\mathbf{g}(\cdot)$  such that  $\int_{\Omega} \mathbf{g}(x) dx = 1$  and  $\mathbf{g}(\cdot)$  approximates the function  $\mathbf{f}(\cdot)$ , then we can compute the integral of Equation 12 as

$$\theta = \int_{\Omega} \mathbf{f}(x) dx = \int_{\Omega} \frac{\mathbf{f}(x)}{\mathbf{g}(x)} \mathbf{g}(x) dx \quad (20)$$

We can then use  $\mathbf{g}(\cdot)$  as a probability density function for the point selection from the domain  $\Omega$  and evaluate  $\theta_N$  as per the details given in Section II-B as,

$$\theta_N = \frac{\sum_{i=1}^N \mathbf{f}(x_i) / \mathbf{g}(x_i)}{N}. \quad (21)$$

<sup>3</sup> Note that, in order to ensure statistical consistency all the error bars for a particular capacitance should intersect with very high probability.

<sup>2</sup> Note that this is equivalent to the Monte Carlo integration scheme explained in Equation 16

Since  $\mathbf{g}(\cdot)$  approximates  $\mathbf{f}(\cdot)$ , most of the samples,  $\mathbf{f}(x_i)/\mathbf{g}(x_i)$  used in the RHS of above summation are close to identity. Thus the variance computed for the samples is much lesser than the one computed using uniform sampling and using Equation 14. Computing such a perfect  $\mathbf{g}(\cdot)$  however is not possible<sup>4</sup>. We need an easy to compute  $\mathbf{g}(\cdot)$  which smoothes out the function  $\mathbf{f}/\mathbf{g}(\cdot)$ . In our case, function  $\mathbf{f}(\cdot)$  is dot product of electric field  $\hat{\mathbf{E}}$  and normal vector  $\hat{\eta}$ . Also, we know that the field  $\hat{\mathbf{E}}$  diminishes with inverse of distance. We choose  $\mathbf{g}(x)$  as  $A/D_x$ , where  $A$  is a constant and  $D_x$  is the distance of point  $x$  from the conductor inside the integration surface  $\Omega$  (see Figure 1 for illustration of  $\hat{\eta}$ ,  $x$  and  $D_x$ ). We choose  $A$  such that,

$$\int_{\Omega} \frac{A}{D_x} dx = 1. \quad (22)$$

### C. Stratified Sampling

The smoothing function  $\mathbf{g}(\cdot)$  used in importance sampling in above section still leaves  $\mathbf{f}/\mathbf{g}(\cdot)$  to have lot of variations. In this section we consider a domain decomposition scheme called ‘stratified sampling’ which can greatly reduce the variance. In stratified sampling, we break the domain of integration into  $k$  disjoint pieces (also called ‘strata’) and apply Monte Carlo sampling to each pieces separately. In Equation 12, if  $\Omega$  is decomposed as  $\Omega = \biguplus_{j=1}^k \Omega_j$ , then an estimator  $\theta_N$  can be computed as,

$$\theta_N = \sum_{j=1}^k \theta_j = \sum_{j=1}^k \left\{ |\Omega_j| \frac{1}{n_j} \sum_{i=1}^{n_j} \mathbf{f}_j(x_i) \right\}, \quad (23)$$

where  $|\Omega_j| = \int_{\Omega_j} dx$ , and  $\mathbf{f}_j(x_i)$  is the value of function  $\mathbf{f}(\cdot)$  at the point  $x_i \in \Omega_j$ . We can decide  $n_j$  beforehand. In our case we keep  $n_j$  proportional to  $|\Omega_j|$ . The variance  $\sigma_N^2$  is also computed piecewise as follows,

$$\sigma_N^2 = \sum_{j=1}^k \sigma_j^2 = \sum_{j=1}^k \left\{ \frac{|\Omega_j|^2 \sum_{i=1}^{n_j} (\mathbf{f}_j(x_i))^2 - n_j \theta_j^2}{n_j - 1} \right\}. \quad (24)$$

The stratification is *well carried out* if variation within the  $\theta_j$  samples is less than the variation in  $\mathbf{f}(\cdot)$  samples as a whole. In our case, an obvious stratification would be to decompose  $\Gamma$  in Equation 2 as  $\Gamma = \biguplus_{j=1}^k \Gamma_j$ . However a well carried out domain decomposition of  $\Gamma$  is not easy to compute a-priori, as this requires knowledge of electric field on the surface. If we had that knowledge we would have computed the integral analytically. Let us look into the Monte Carlo integration process for Equation 2. The samples using which an estimate  $Q_N$  is computed are  $q_i = A_{C_i} \epsilon_{p_i} \hat{\mathbf{G}}_{p_i}(p_j) \cdot \hat{\eta}_{p_i}$ . A sample  $q_i$  is computed by evaluating dot product of electric field Green’s function  $\hat{\mathbf{G}}_{p_i}(\cdot)$  and the normal vector  $\hat{\eta}_{p_i}$  at a point  $p_j$  randomly selected on the cube centered at point  $p_i$ . Finally this dot-product is normalized with the cube’s surface area  $A_{C_i}$  and the point epsilon value  $\epsilon_{p_i}$ . If we consider cubes of unit area in uniform dielectric medium, then the sample value  $q_i$  is completely determined by the dot product. In Figure 5 we plot such a dot product on six faces of a cube with uniform dielectric inside it. We note that the dot product on top and bottom faces form a bell shaped curve, whereas the dot-product on other 4 faces form twin cusp surfaces with positive values on one half of the face and negative on the other. If we take values of dot product on points on the surface of a cube we get a lot of variation as per the plots in Figure 5. The aim is to decompose the surface of cube into strata such that this variation is as little as possible within a stratum. We do so by decomposing the surface of cube as per the dot product contour plots. We decompose top and bottom faces into 7 concentric square rings (see Figure 4(A)). We

decompose the union of 4 faces into two pieces, one corresponding to positive cusp shape and other corresponding to the negative cusp shape. This totals to be  $7+7+2=16$  pieces of the cube’s surface<sup>5</sup>. We then apply this decomposition onto the domain of integral in Equation 7. The domain out there can be thought of as  $\Gamma \times C_p$ . Its a cross product of points  $p$  on  $\Gamma$  and points on cube’s surface  $C_p$  centered at each point  $p$ . The domain decomposition is done as follows,

$$\Gamma \times C_p = \bigoplus_{i=1}^{16} \Gamma \times C_p^i, \quad (25)$$

where  $C_p^i$  for  $i = 1, \dots, 16$  correspond to 16-way stratification of cube’s surface as described above. We collect the samples in separate strata as per their origin in the domain. We compute mean and variance for  $Q_N$  by using Equations 23 and 24. In Figure 6 we plot the conver-

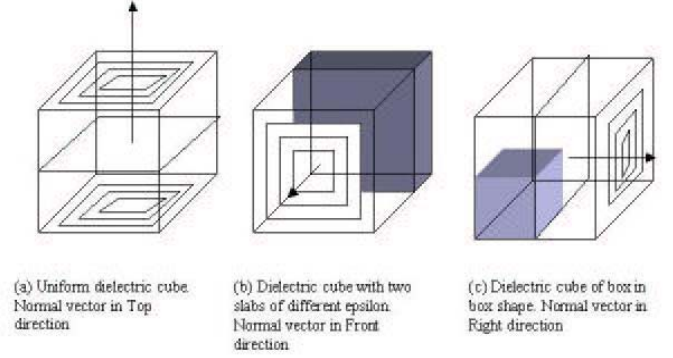


Fig. 4. Marking of cube decomposition in three different cases.

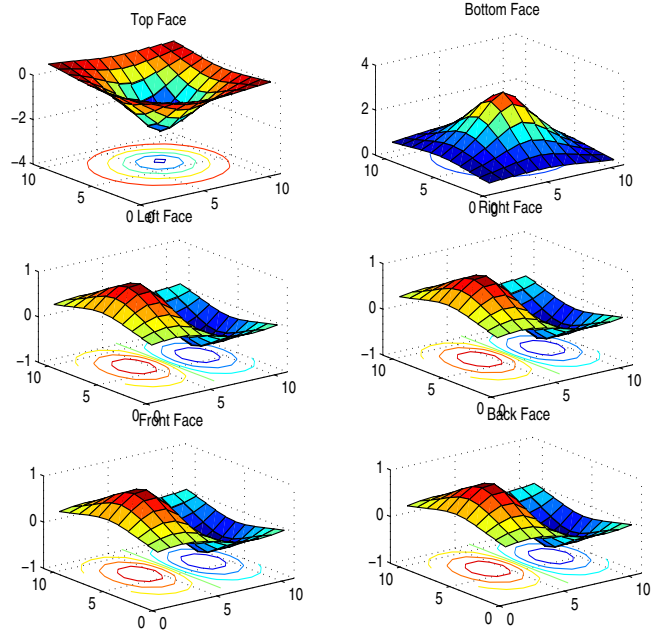


Fig. 5. Contour plots of dot product of normal vector and electric field Green’s function on 6 possible faces of a uniform dielectric cube as shown in Figure 4(A).

gence behavior of the same net with importance sampling and stratified sampling used during Monte Carlo integration. Note that the errorbars are much smaller than the ones in Figure 3. In Section V we quantify these improvements using more industry designs.

<sup>4</sup> Otherwise we would have done an analytical integration!

<sup>5</sup> In Figure 4, for ease of drawing, we have only shown 4 concentric square rings.

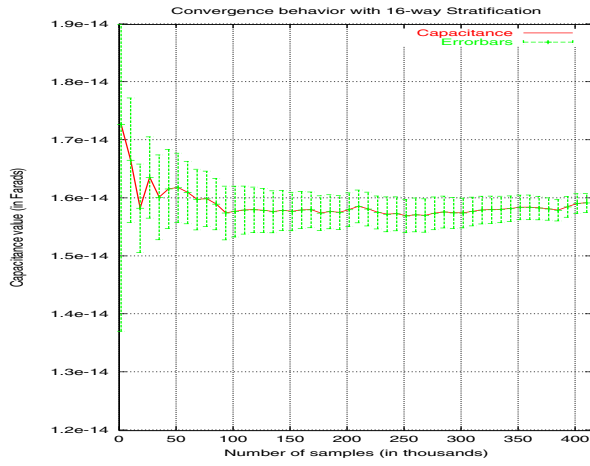


Fig. 6. Convergence behavior of a self cap with 16-way stratification.

### C.1 Extension to Non-Uniform Case

In modern technologies we need to perform random walks in complex dielectric environments. The space is well represented as slabs of different dielectric strengths, and there are conformal dielectric objects around the conductors. We then encounter non-uniform dielectric cubes around the point on the Gaussian surface. The contour plots for dot product of electric field and normal vector on such dielectric cubes are different from the one shown in Figure 5. In Figure 4(B) and (C) we show two different dielectric cubes, one having two slabs of different dielectric strength, and other having a small dielectric box of different dielectric strength inside the bigger unit dielectric cube (referred to as *box-in-box* case). Note that the direction of the normal vectors are also different<sup>6</sup>. The corresponding dot product contours are shown in Figure 7 and 8 respectively. We can note that the cube stratification (decomposition of cube's surface into pieces or stratas) build from uniform dielectric cube still works good for slab case. For conformal case its not as good as desired. The dot product undergoes more variance within each strata of conformal cube as compared to the uniform cube. But since box-in-box case is seldom encountered in random walk process and can often be avoided, the cube stratification built with uniform dielectric cube works well in practice.

### IV. MODIFIED RANDOM WALK ALGORITHM

In this section, we briefly touch upon the modifications to random walk algorithm given in [8].

1. Build rectilinear Gaussian surface around the nets to be extracted.
2. Pick a point on a Gaussian surface as per the importance sampling scheme described in Section III-B.
3. Build a maximal conductor free cube centered at this point.
4. Perform cube stratification as per the normal vector and its dot product with electric field Green's function for the uniform cube.
5. Pick a point on the surface of this cube.
6. Get the strata-id  $j$  for the point as per its location on the cube.
7. If the point is on surface of any other conductor, the walk ends.
8. If the point is in dielectric space, build another conductor free cube.
9. Use its electric potential Green's function as pdf to select a point on this cube. If the point is on the surface of conductor the walk ends, otherwise go to previous step.
10. Once the walks ends, register the weight  $q_i$  into the strata  $j$ .
11. Compute capacitance estimate and variance using Equations 23 and 24.

<sup>6</sup>In 3D space, with rectilinear Gaussian surface there are only 6 possible directions to normal vectors.

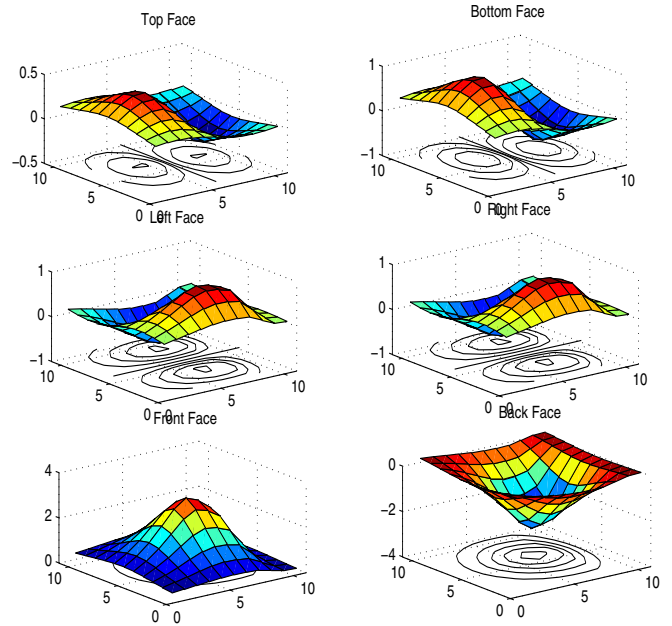


Fig. 7. Contour plots of dot product of normal vector and electric field Green's function on 6 possible faces of a cube with two slabs of different dielectric strengths as shown in Figure 4(B).

### V. EXPERIMENTAL RESULTS

In this section, we present experimental results to quantify the run-time improvements with the usage of importance and stratified sampling described in Section III. We report run-times to extract self capacitance for three largest nets in 15 different designs. Each self capacitance is extracted to be within 3% of accurate value. The design specifications are given in Table I. In column 5-8 of the same table we report the runtime for extraction on a 2.4GHz linux machine with 4GB RAM. Base case corresponds to the run-times with equi-distant Gaussian surfaces, IS case corresponds to the run-times observed with use of importance sampling alone and finally SS case corresponds to the run-times observed with employing stratified sampling on top of importance sampling. In column 8 we report the total speed-up observed with both these variance reduction methods. In the bottom we report the total run times and speedup for extraction of all the 15 designs (3 nets in each design). Note that we get overall speedup of 6.9X to 44X with these variance report methods. Looking at the total numbers, we get a speedup of 1.6X using just the importance sampling, and another 10X with stratified sampling, resulting into average speedup of 16X. All these designs were in uniform dielectric technology. We migrated these designs to a more realistic technology having inter-layer dielectrics etc. In that case we get overall speedups between 3.5X and 11X. On an average we get speedup of about 1.4X with importance sampling and another 5.2X with stratified sampling, resulting into overall average speedup of about 7.3X. We further migrated these designs to technologies having conformal dielectrics on most of the conducting layers. We get overall speedups between 2.3X and 6X. We get about 1.4X speedup with importance sampling, and 3.3X with stratified sampling, resulting into average speedup of about 4.6X.

Note that we get a speedup of about 1.4X-1.6X with use of importance sampling. This is true for all technologies. However, the benefits of stratified sampling are maximum in uniform dielectric technologies. The reason for this is the fact that the stratification scheme (decomposition of cube's surface as per the contour plots of dot product of electric field and normal vector) is derived from plots for uniform dielectric Green's function. This is the most commonly used Green's function

TABLE I

SPECIFICATION OF NUMBER OF METAL LAYERS (L), TOTAL NUMBER OF BOXES (B) AND TOTAL NUMBER OF NETS (N) IN THE TEST DESIGNS. RUNTIMES IN SECONDS TO PERFORM RANDOM WALKS IN UNIFORM, SLABS AND CONFORMAL DIELECTRIC CONFIGURATIONS. BASE IS WITHOUT ANY VARIANCE REDUCTION SCHEME WITH EQUI-DISTANT GAUSSIAN SURFACE, IS IS WITH IMPORTANCE SAMPLING AND SS IS WITH STRATIFIED SAMPLING. SPEEDUP IS OVERALL IMPROVEMENT FACTOR WITH USAGE OF BOTH IS AND SS.

| Design  | L | B     | N      | Uniform |      |     |         | Slabs |      |      |         | Conformal |      |      |         |
|---------|---|-------|--------|---------|------|-----|---------|-------|------|------|---------|-----------|------|------|---------|
|         |   |       |        | Base    | IS   | SS  | Speedup | Base  | IS   | SS   | Speedup | Base      | IS   | SS   | Speedup |
| des1    | 5 | 0.29M | 7.2k   | 39      | 25   | 2   | 19.5X   | 155   | 106  | 16   | 9.6X    | 384       | 259  | 70   | 5.4X    |
| des2    | 5 | 1.76M | 41.7k  | 47      | 28   | 2   | 23.5X   | 136   | 101  | 18   | 7.5X    | 317       | 279  | 75   | 4.2X    |
| des3    | 5 | 4.36M | 107.7k | 48      | 31   | 2   | 24.0X   | 165   | 121  | 21   | 7.8X    | 441       | 382  | 112  | 3.9X    |
| des4    | 5 | 0.20M | 4.3k   | 37      | 21   | 1   | 37.0X   | 123   | 83   | 16   | 7.6X    | 358       | 259  | 79   | 4.8X    |
| des5    | 3 | 0.28M | 7.5k   | 74      | 44   | 7   | 10.5X   | 64    | 44   | 10   | 6.4X    | 532       | 266  | 108  | 4.9X    |
| des6    | 5 | 4.28M | 108.4k | 43      | 28   | 2   | 21.5X   | 174   | 134  | 24   | 7.2X    | 483       | 405  | 108  | 4.4X    |
| des7    | 3 | 1.83M | 44.3k  | 36      | 18   | 3   | 12.0X   | 67    | 63   | 19   | 3.5X    | 174       | 163  | 73   | 2.3X    |
| des8    | 4 | 3.83M | 115.9k | 76      | 52   | 11  | 6.9X    | 88    | 49   | 12   | 7.3X    | 449       | 197  | 78   | 5.7X    |
| des9    | 6 | 6.84M | 148.3k | 71      | 53   | 6   | 11.8X   | 109   | 44   | 10   | 10.9X   | 248       | 130  | 43   | 5.7X    |
| des10   | 4 | 6.38M | 181.8k | 45      | 35   | 2   | 22.5X   | 164   | 116  | 15   | 10.9X   | 313       | 229  | 53   | 5.9X    |
| des11   | 4 | 0.56M | 14.0k  | 67      | 40   | 3   | 22.3X   | 153   | 126  | 32   | 4.7X    | 306       | 283  | 89   | 3.4X    |
| des12   | 3 | 1.82M | 45.8k  | 196     | 107  | 10  | 19.6X   | 99    | 54   | 13   | 7.6X    | 425       | 226  | 79   | 5.3X    |
| des13   | 5 | 0.24M | 5.3k   | 51      | 26   | 2   | 25.5X   | 122   | 86   | 13   | 9.3X    | 311       | 200  | 55   | 5.6X    |
| des14   | 5 | 1.07M | 24.1k  | 44      | 22   | 1   | 44.0X   | 125   | 96   | 13   | 9.6X    | 397       | 300  | 83   | 4.7X    |
| des15   | 6 | 6.16M | 150.2k | 65      | 50   | 4   | 16.2X   | 175   | 147  | 28   | 6.2X    | 366       | 295  | 95   | 3.8X    |
| Total   |   |       |        | 939     | 580  | 58  |         | 1919  | 1370 | 260  |         | 5504      | 3873 | 1200 |         |
| Speedup |   |       |        |         | 1.6X | 10X |         |       | 1.4X | 5.2X |         |           | 1.4X | 3.2X |         |

with interlayer dielectrics and speedup of 4.6X is observed for technologies having conformal dielectrics. To the best of our knowledge this is the first reporting of variance reduction techniques to effectively reduce runtime of Monte Carlo capacitance extraction.

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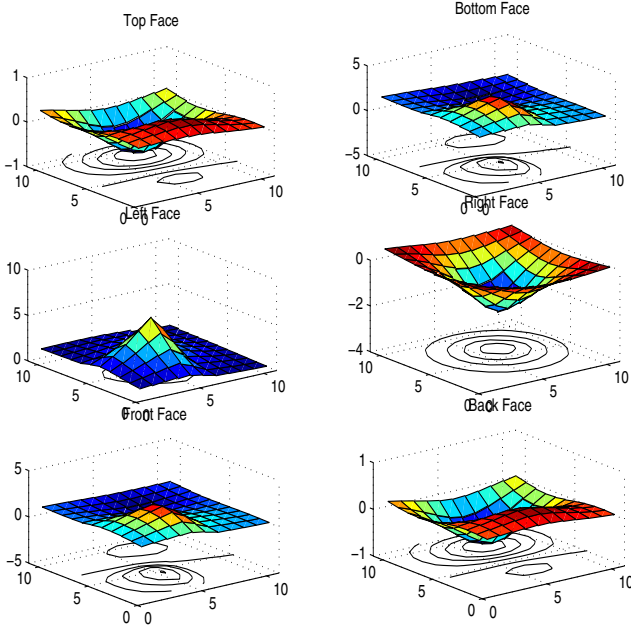


Fig. 8. Contour plots of dot product of normal vector and electric field Green's function on 6 possible faces of a cube with box-in-box configuration as shown in Figure 4(C).

in random walk process. We need to choose the stratification scheme up-front, i.e., before the random walk process, and we can choose only one stratification scheme. Choosing cube's decomposition as per the contour plots for other Green's functions gives worse overall results.

## VI. CONCLUSIONS

We have presented an efficient implementation of Monte Carlo algorithm for capacitance extraction. Variance reduction techniques are applied to the statistical evaluation of charge integral. Average speedup of 16X is observed in capacitance extraction problems with uniform dielectric technologies. Speedup of 7.3X is observed for technologies